Python for Data Science

Learn Python programming and apply it to real-world data analysis and visualization tasks.

Welcome to the course! This document contains all the materials, lessons, and exercises you will need to successfully complete the curriculum. We recommend reading through this guide carefully and taking notes as you progress.

Category: Data | Level: Intermediate | Est. Time: 10 Hours

Course Objectives & Overview

By the end of this course, you will have a solid understanding of the core concepts, practical applications, and best practices associated with this topic. This structured material is designed to guide you step-by-step from foundational theories to advanced problem-solving techniques.

Ensure you complete all practical exercises and review the lesson summaries before attempting the final exam.

Lesson 1: Python Fundamentals

Python is a high-level, interpreted programming language known for its clear syntax.

• Installing Python and setting up a virtual environment
• Variables and data types: int, float, str, bool, list, dict, tuple, set
• Operators and expressions
• String formatting: f-strings, .format()
• Conditional statements: if, elif, else
• Loops: for, while, list comprehensions
• Functions: def, parameters, return, *args, **kwargs
• Modules and imports
• File I/O: open(), read(), write()
• Exception handling: try/except/finally

Key concept: Python uses indentation (whitespace) to define code blocks — there are no curly braces.

Key Takeaways:

• Review the main concepts outlined above.
• Practice the techniques discussed.
• Write down any questions for further research.

Lesson 2: NumPy & Pandas

NumPy and Pandas are the core libraries for data science in Python.

NumPy:
• Arrays: np.array(), shape, dtype
• Array operations: slicing, indexing, broadcasting
• Math functions: np.sum(), np.mean(), np.std()

Pandas:
• Series and DataFrame
• Reading data: pd.read_csv(), pd.read_excel()
• Exploring data: .head(), .info(), .describe()
• Selecting data: loc, iloc, boolean indexing
• Handling missing data: .isnull(), .dropna(), .fillna()
• Grouping and aggregating: .groupby(), .agg()

Key concept: A DataFrame is like a spreadsheet in Python — rows are observations, columns are variables.

Key Takeaways:

• Review the main concepts outlined above.
• Practice the techniques discussed.
• Write down any questions for further research.

Conclusion & Next Steps

Congratulations on reaching the end of the written material for this course! Your next step is to log into the platform and take the final assessment. Review your personal notes, ensure you understand the key takeaways from each lesson, and apply what you have learned.

Good luck!